Week-2

Introduction to Database Systems

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How is a house constructed?

- If you want to build a house what steps will you follow?
  - We will give our requirements to the architect.
  - He will show us some paperwork i.e. a map or a 3d model.
  - Then the foundations are laid and grey structure is built according to the preferred material.
  - We then finalize about the finishing and furnishing.

Design Phases of Database

1. Requirement Definition and analysis

- Design of a new application for an existing database

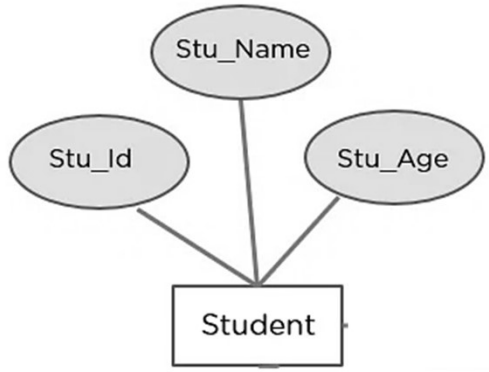
  OR

- Design of a new database

2. Conceptual Design

- These requirements are documented in detail to form a conceptual design
- Conceptual design can be represented and manipulated using some computerized tool so that it can easily be maintained, modified and transformed to database implementation.

  Example: Entity Relationship Model



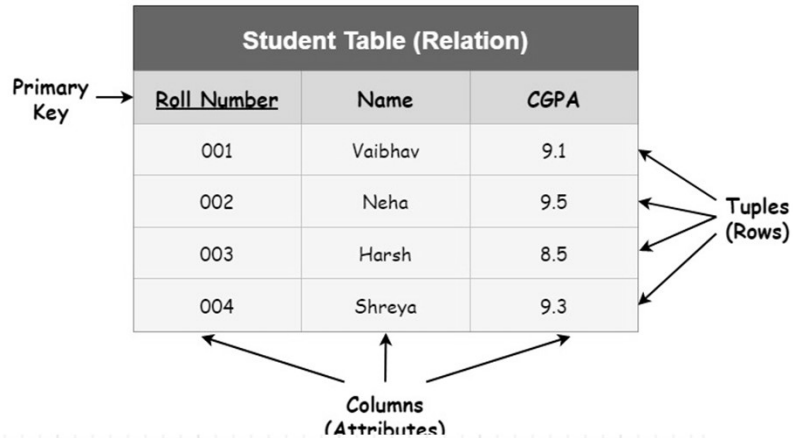
Design Phases of Database

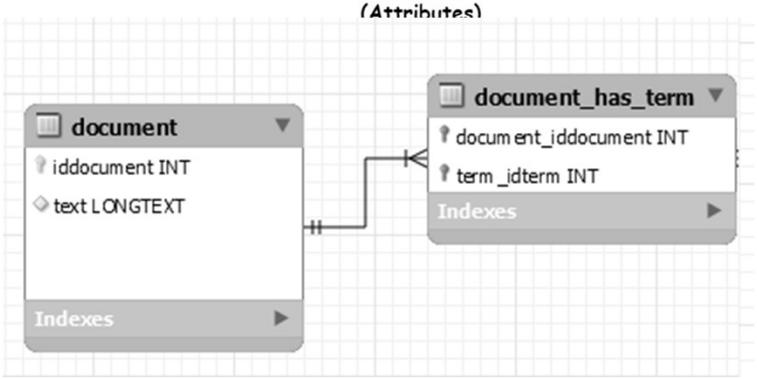
3. Logical Design

- Conceptual design is transformed into logical design
- Logical design can be expressed in a **data model**
- Example: Relational Data Model

4. Physical Design

- In this stage further specifications are provided for storing and accessing the database.
- Database design is implemented , populated with actual data and continuously maintained to reflect the actual state of miniworld.





Entity

Entity in Database

- An entity is an object that exists. It doesn't have to do anything; it just has to exist. In database administration, an entity can be a single thing, person, place, or object. Data can be stored about such entities.
- **Strong Entity**

  An entity is strong if it can independently exist

  Example: Student is a strong entity type as it has a unique ID for each student.
- **Weak Entity**

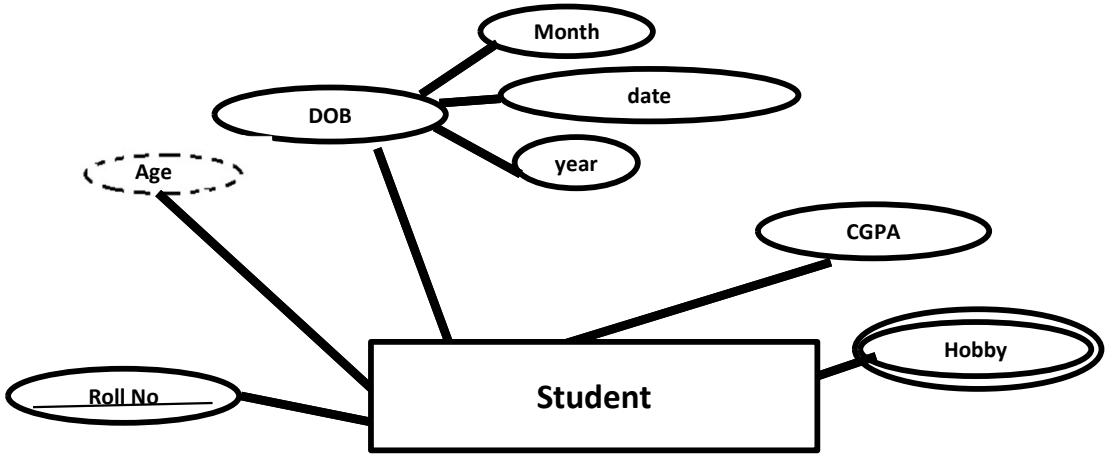
  An entity is weak when its existence depends on another entity.

  Example: Section can't exist without class

Attributes of an Entity

- An attribute is a property or characteristic of an entity.
- An entity may contain any number of attributes.
- One of the attributes is considered as the primary key.

Example:
a customer can have attributes such as name, email, and address.



Entity Set

An entity set is a collection or set of all entities of a particular entity type at any point in time. The type of all the entities should be the same.



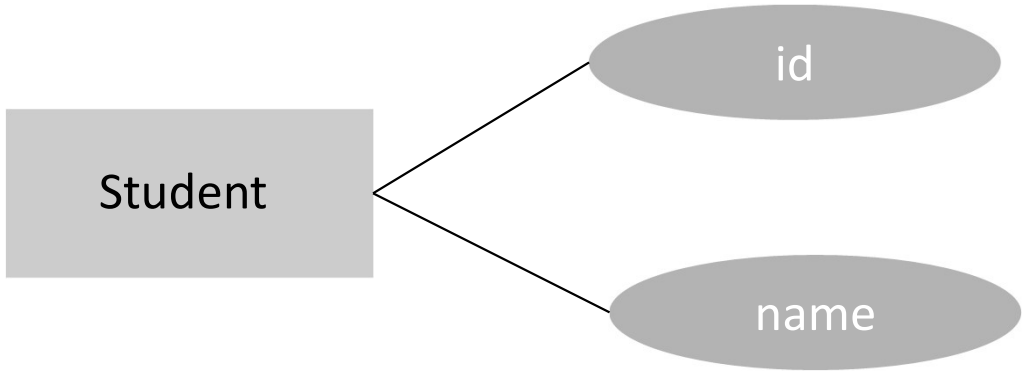
Entity Set

Example :

- The collection of all the students from the student table at a particular instant of time is an example of an entity set.
- The collection of all the employees from the employee table at a particular instant of time is an example of an entity set.

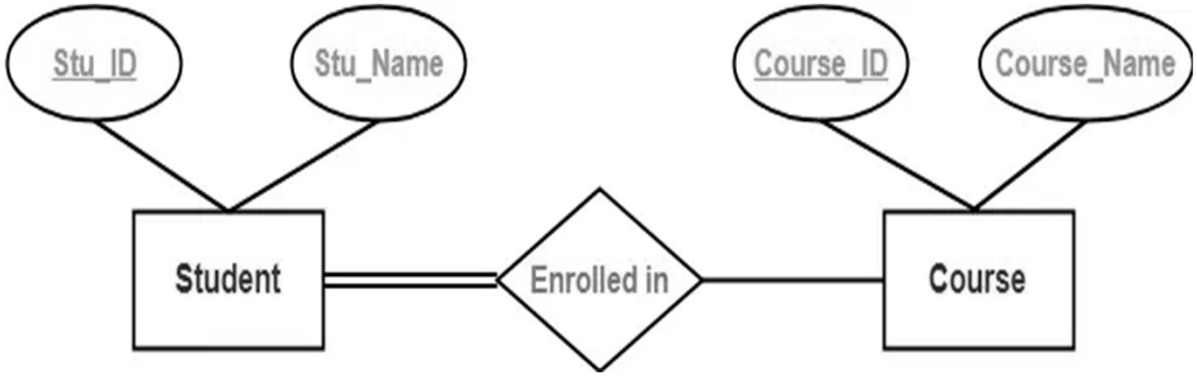
Entity Relation Diagram

- In an Entity-Relation model, attributes are represented in an elliptical shape.
- Properties of an entity



Entity Relation Diagram

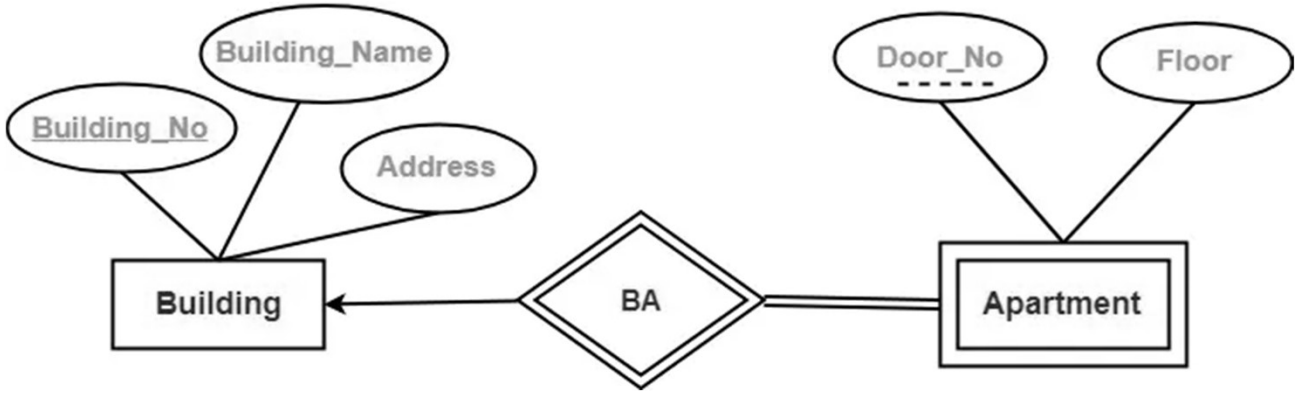
Symbols Used for Strong Entity:
•A single rectangle is used for representing a strong entity set.
•A diamond symbol is used for representing the relationship that exists between two strong entity sets.



In this ER diagram:
•Two strong entity sets "**Student**" and "**Course**" are related to each other.
•Student ID and Student name are the attributes of entity set "Student".
•Student ID is the primary key using which any student can be identified uniquely.
•Course ID and Course name are the attributes of entity set "Course".
•Course ID is the primary key using which any course can be identified uniquely.
•Double line between Student and relationship set signifies total participation.
•It suggests that each student must be enrolled in at least one course.
•Single line between Course and relationship set signifies partial participation.
•It suggests that there might exist some courses for which no enrollments are made.

Entity Relation Diagram

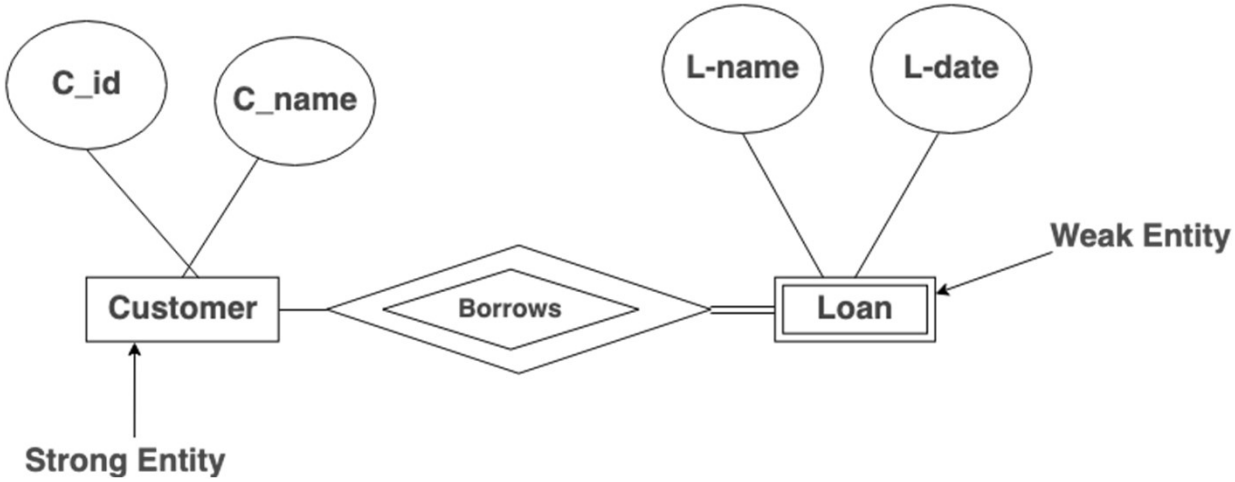
Symbols Used for Weak Entity:
•A double rectangle is used for representing a weak entity set.
•A double diamond symbol is used for representing the relationship that exists between the strong and weak entity sets and this relationship is known as **identifying relationship**.
•A double line is used for representing the connection of the weak entity set with the relationship set.
•Total participation always exists in the identifying relationship.



In this ER diagram,
•One strong entity set "**Building**" and one weak entity set "**Apartment**" are related to each other.
•Strong entity set "Building" has building number as its primary key.
•Door number is the discriminator of the weak entity set "Apartment".
•This is because door number alone can not identify an apartment uniquely as there may be several other buildings having the same door number.
•Double line between Apartment and relationship set signifies total participation.
•It suggests that each apartment must be present in at least one building.
•Single line between Building and relationship set signifies partial participation.
•It suggests that there might exist some buildings which has no apartment.

Entity Relation Diagram

Symbols Used for Weak Entity:
•A double rectangle is used for representing a weak entity set.
•A **double diamond symbol** is used for representing the relationship that exists between the strong and weak entity sets and this relationship is known as **identifying relationship**.
•**A double line** is used for representing the connection of the weak entity set with the relationship set.
•Total participation always exists in the identifying relationship.



Weak entities are represented with **double rectangular** box in the ER Diagram and the identifying relationships are represented with double diamond. Partial Key attributes are represented with dotted lines.

Types of an attribute

Types of attributes

- Key attribute [Student ID, CNIC, Passport Number]
- Derived attribute [Age can be derived from DOB]
- Simple attribute [Employee ID]
- Composite attribute [The address can be further split into house number, street number, city, state etc]
- Multivalued attribute [Contact number]

Multivalued key vs Composite Key

Multivalued Attribute:
Imagine a database for a library. Each book can have multiple authors, and each author can write multiple books. Here, "authors" would be a multivalued attribute for the entity "book" because each book can have multiple authors. Similarly, "books" would be a multivalued attribute for the entity "author" because each author can write multiple books.
Example:

- Book entity: Title, ISBN, Genre, Authors (multivalued attribute)
    - Example of a book: "Introduction to Database Systems," 978-0134401885, Computer Science, [C.J. Date, Hugh Darwen]

Composite Attribute:
Composite attributes are attributes that can be divided into smaller sub-parts, each representing a simpler attribute. These simpler attributes together form the composite attribute.
Example:

- Address entity: Street Address, City, State, Zip Code (composite attribute)
    - Example of an address: 123 Main St, Anytown, CA, 12345

Keys and its Types

- A key in DBMS is an attribute or a set of attributes that help to uniquely identify a tuple (or row) in a relation (or table).
- Keys are also used to establish relationships between the different tables and columns of a relational database.
- Individual values in a key are called key values.

Roll No.	Name	Age	Gpa
1	Aryan	21	3
2	Sachin	25	4
3	Prince	20	2.5
4	Anuj	21	3.5

Keys and its Types

1. **Primary Key**
2. **Candidate Key**
3. **Super Key**
4. **Alternate Key**
5. **Foreign Key**
6. **Composite Key**
7. **Unique Key**

Keys and its Types

Primary Key

A primary key, also called a primary keyword, is a column in a relational database table that's distinctive for each record.

It's a unique identifier, such as a :

- driver's license number, CNIC.

Primary Key ↑

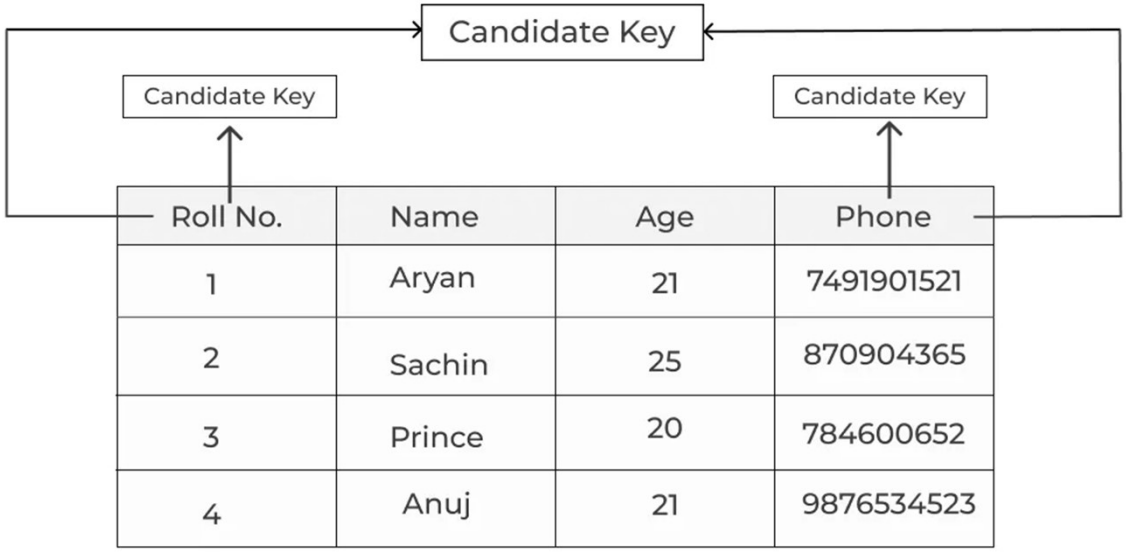
Roll No.	Name	Age	Gpa
1	Aryan	21	3
2	Sachin	25	4
3	Prince	20	2.5
4	Anuj	21	3.5

Keys and its Types

Candidate Key

For example, employee relation has two attributes, Employee_Id and NTN, that act as candidate keys.

In this relation, Employee_Id is chosen as the primary key, so the other candidate key, NTN, acts as the Alternate key.

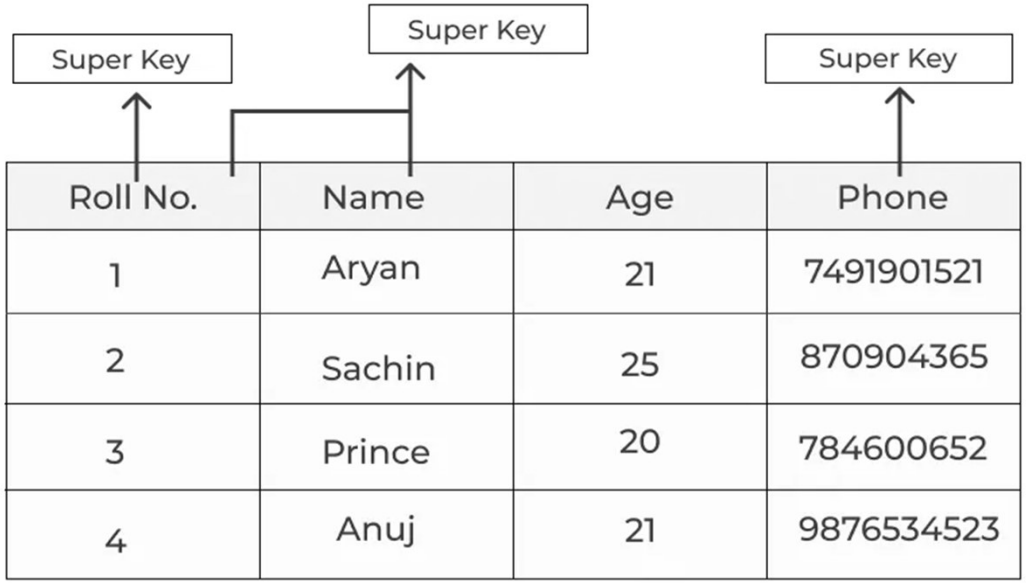


Roll No.	Name	Age	Phone
1	Aryan	21	7491901521
2	Sachin	25	870904365
3	Prince	20	784600652
4	Anuj	21	9876534523

Keys and its Types

Super Key

- If you add any other column/attribute to a Primary Key then it become a super key.
- E.g. RollNo + Name is a Super Key.
- A Super key may have additional attributes not needed for unique identification.

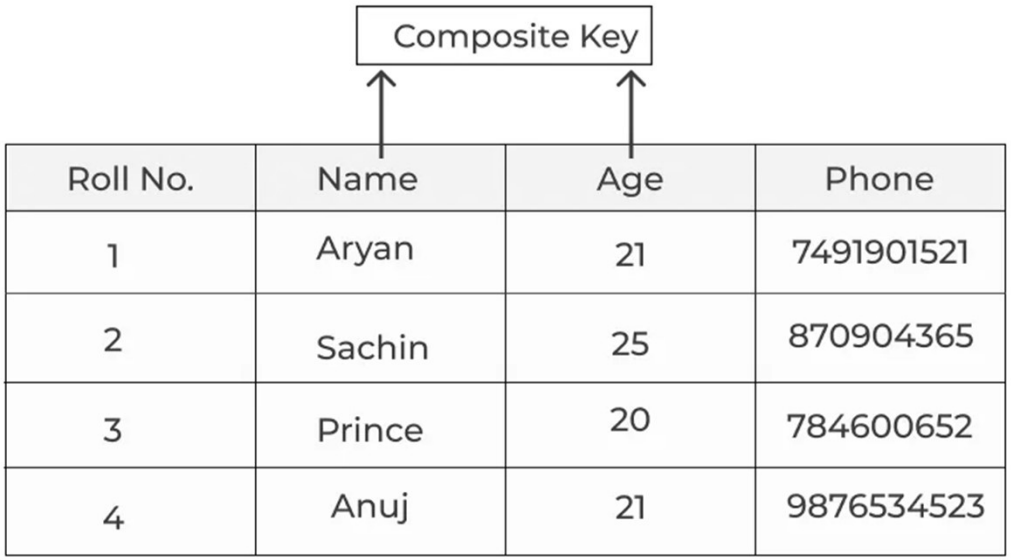


Roll No.	Name	Age	Phone
1	Aryan	21	7491901521
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3	Prince	20	784600652
4	Anuj	21	9876534523

Keys and its Types

Composite Key

- A Composite Key is a combination of one or more attributes. If a single column alone fails to be served as a primary key then combination of columns would help to uniquely access a record from table such type of keys are nothing but composite keys.



Roll No.	Name	Age	Phone
1	Aryan	21	7491901521
2	Sachin	25	870904365
3	Prince	20	784600652
4	Anuj	21	9876534523

Keys and its Types

Alternate Key

An alternate key is a column or group of columns that uniquely identify every row in a table.

If a table has more than one candidate key, one of them will become the primary key and rest of all are called alternate keys.

Keys and its Types

Foreign Key

A foreign key is a column or group of columns in a relational database table that provides a link between data in two tables. It is a column (or columns) that references a column (most often the primary key) of another table.

Student_Details:

Roll No.	Name	Course_Id → Key
1	Aryan	101
2	Sachin	102
3	Prince	103

Student_Marks:

Foreign Key ←

Course_Id	Gpa
101	3
102	4
103	2.5

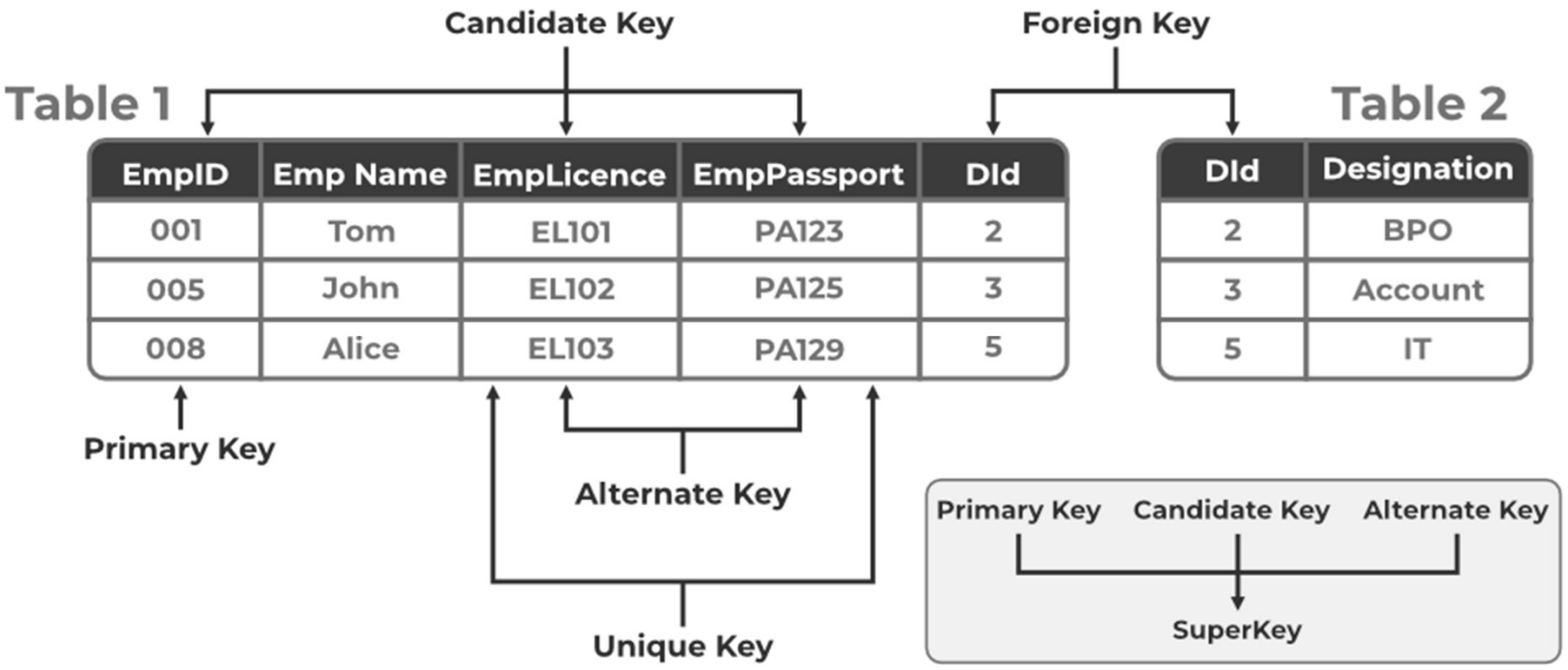
Keys and its Types

Unique Key

Each employee must have a unique number; hence, the EmployeeNumber column implements the UNIQUE constraint.

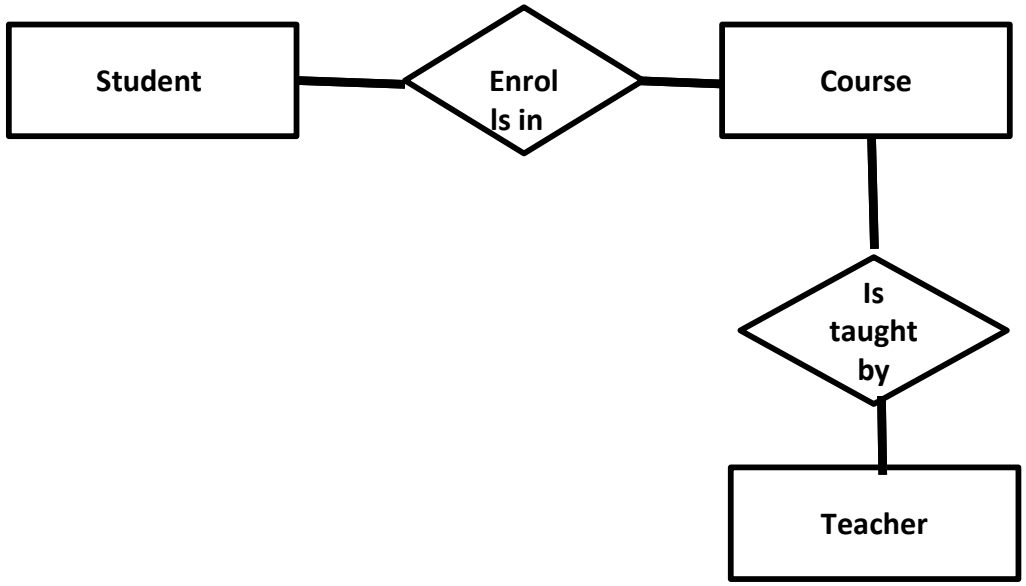
By doing so, we ensure that no employees in the same company have the same employee number.

Keys and its Types



Relationship representation in ERD

Please note: Readability is Left to Right & Top to bottom



Constraints on Relationship

Cardinality

- One to many (1:N)
- One to one (1:1)
- Many to many (M:N)

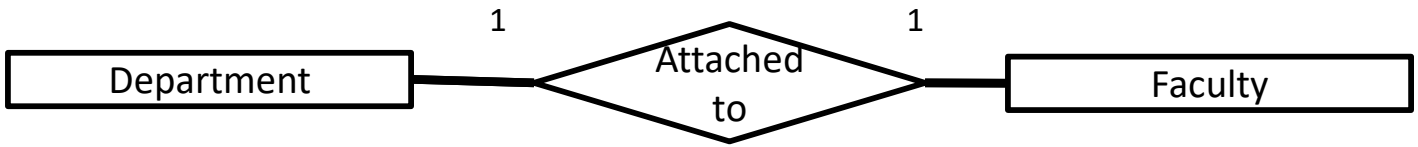
Participation

- Total Participation
- Partial participation

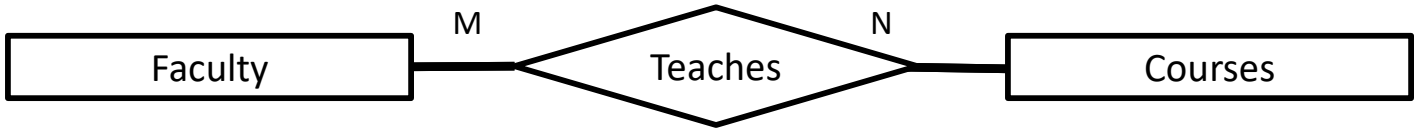
Relationship: cardinality

Cardinality

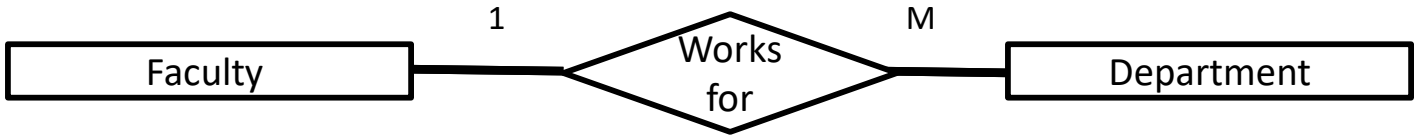
- 1 : M



- M : N

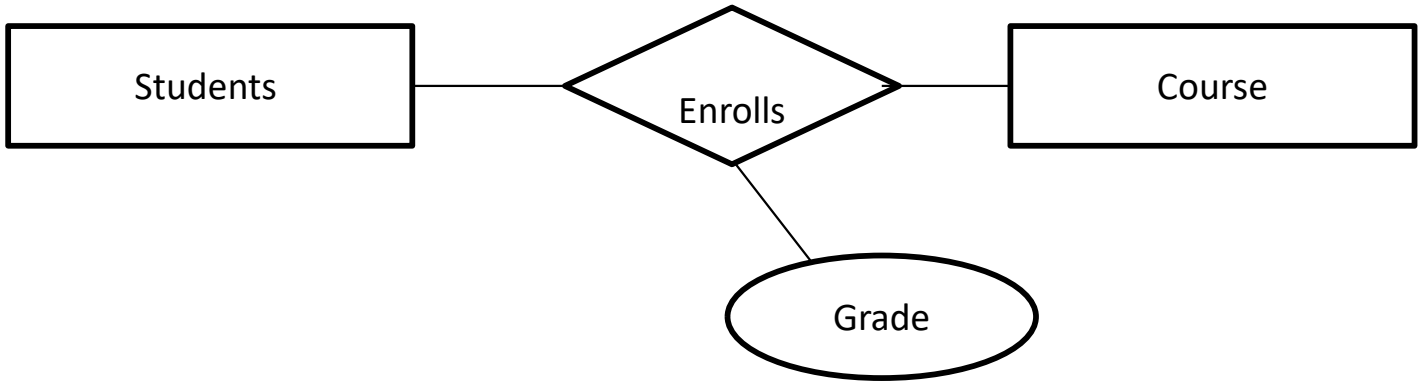


- 1 : 1



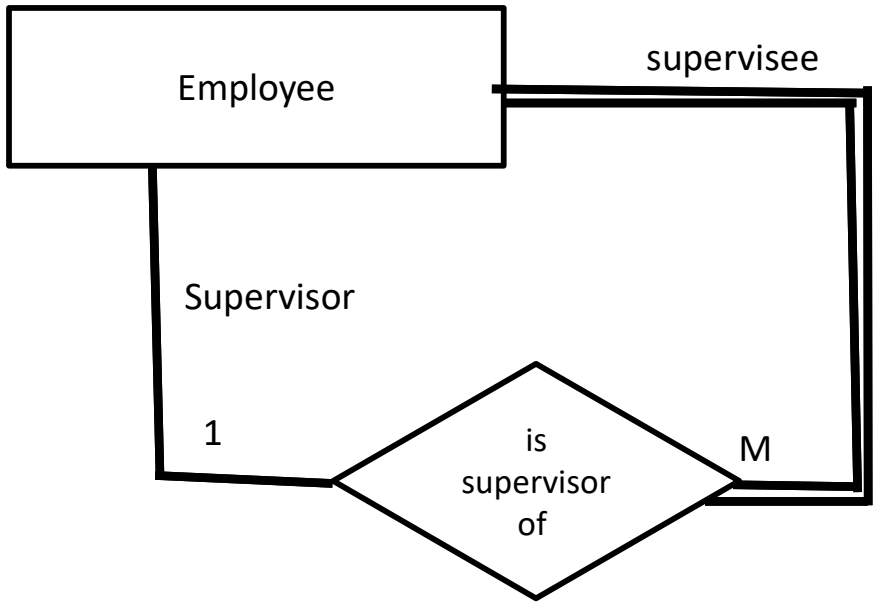
Attributes of relationship

- A relationship can have attributes:
  - For example, grade of enrolls
  - Its value for each relationship instance describes the grade a student takes when he is enrolled in a course.
    - A value of Grade depends on a particular (Student, Course) combination

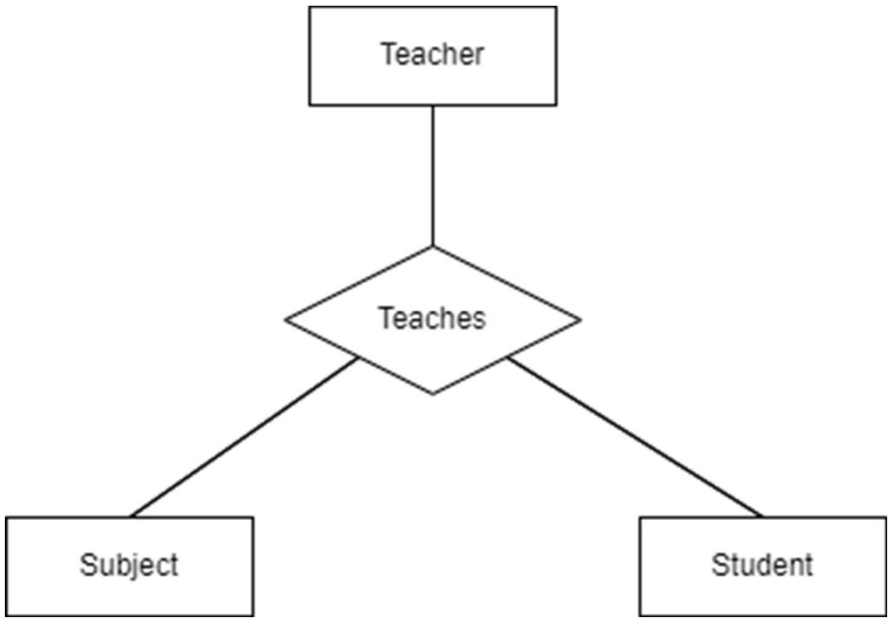


Recursive Relationship: Rolenames

- Employee is supervisor of other employees



Ternary relationship



Thank You all!